Practical - 11

Practical Title: Student Management CRUD with Search and Filter

Problem Definition: Create a student management module to add, view, edit, delete, search and filter student records. All database operations must use MySQLi statements, prepared statements, and clear success/failure messages.

A. Theory & Analysis

1. Are SELECT, INSERT, UPDATE, and DELETE operations working?

Yes. All four CRUD operations are implemented with prepared statements: SELECT queries fetch student records with search filters; INSERT enrolls new students after duplicate validation; UPDATE modifies student details; and DELETE removes records cleanly.

2. Is search by name/email/course implemented?

Yes. The search query uses SQL LIKE '%keyword%' across enrollment_no, full_name, and email. It also supports exact course filtering from a dropdown menu, allowing combined search and filter criteria.

3. Are delete operations confirmed before execution?

Yes. When the administrator clicks Delete, a confirmation modal dialog opens showing the student name and enrollment number. Deletion executes only upon explicit user confirmation, preventing accidental record removal.

B. Implementation Details

I developed the student management module in pages/manage_students.php with full CRUD operations for 21 student records. It features search by name or enrollment, course filtering, Add Student modal, Edit Student modal, Delete confirmation modal, CSV data export (?export=csv), and audit logging.

C. Evidence of Work

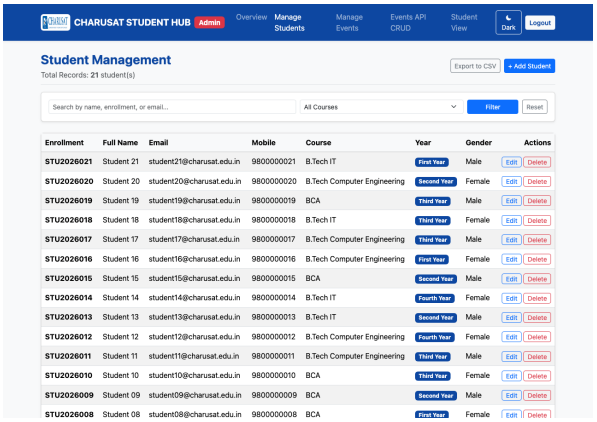


Figure 1: Student Management Administrative Table with Search and Action Modals

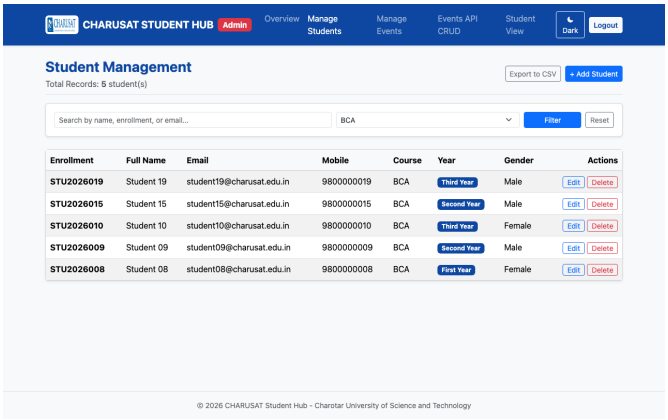


Figure 2: Filtered Student Records by Academic Course with Real-Time Counter

D. Repository Link

GitHub Repository: https://github.com/neel92654/WDF_A_ARP

E. Learning Outcomes

I learned how to construct an administrative database CRUD module in PHP and MySQL using prepared statements, multi-field search, course filtering, confirmation modals, and CSV data export.